

The relationship between plasma carotenoids and depressive symptoms in older persons

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Objective We examined the cross-sectional and longitudinal relationship between plasma carotenoids and depressive symptoms over a six-year follow-up in older persons. **Methods and Materials** This research is part of the InCHIANTI Study, a prospective population-based study of older persons in Tuscany, Italy. The sample for this analysis included 958 women and men aged 65 years and older. Plasma total carotenoids were assessed at baseline. Depressive symptoms were assessed at baseline and at the 3- and 6-year follow-up using the Center for Epidemiological Studies-Depression Scale (CES-D). Depressed mood was defined as CES-D $\geq$ 20. **Results** At baseline, higher total carotenoids level were associated with lower probability of depressed mood (OR=0.82, 95%CI=0.68–0.99, p=0.04) after adjustment for sociodemographic, health and inflammation. After the exclusion of participants with baseline depressed mood and use of antidepressants, higher total carotenoids level were associated with lower risk of incident depressed mood (OR=0.72, 95%CI=0.52–0.99, p=0.04) at 6-year follow-up, after adjustment for confounders plus baseline CES-D. Inflammatory marker Interleukin-1 receptor antagonist partially mediated this association. **Discussion** Low plasma concentrations of carotenoids are associated with depressive symptoms and predict the development of new depressive symptoms in older persons. Understanding the mechanism of this association may reveal potential targets for prevention and treatment.